

Classification

Logistic Regression, Generative Classifiers,
and LDA/QDA

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MATRA LAB

Session Roadmap

Recap & Objectives

Mathematical Prerequisites

Why Linear Regression Fails

The Logistic Model

Estimating the Coefficients

Multiple & Multinomial Logistic Regression

Generative Models: Bayes' Theorem, LDA, QDA

Comparing Classification Methods

Applied View

Summary

SECTION 1

Recap & Objectives

Where We Were: Linear Regression

From the previous session

For a **quantitative** response, least squares gives the closed-form estimator

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y},$$

obtained by minimising the squared-error loss $\text{RSS}(\beta) = (\mathbf{y} - \mathbf{X}\beta)^T (\mathbf{y} - \mathbf{X}\beta)$.

The gap this session fills

What if Y is a *category*, not a number? Squared error and a straight line no longer make sense. We keep the same linear predictor $\mathbf{x}^T \beta$, but pass it through a **link function** that squashes it into $[0, 1]$ — this is **logistic regression**.

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Why Do We Need Classification?

Regression predicts numbers

- ▶ House price = \$250,000
- ▶ Salary = \$80,000
- ▶ Sales = 1200 units

Classification predicts categories

- ▶ Email → Spam / Not Spam
- ▶ Tumour → Benign / Malignant
- ▶ Credit card → Default / No Default
- ▶ Stock market → Up / Down

One-line contrast

Regression → numbers. Classification → categories.

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Learning Objectives

By the end of this session you will be able to:

1. Explain why linear regression fails for a categorical response.
2. Derive the logistic function and interpret its coefficients via odds and log-odds.
3. Estimate logistic-regression coefficients by maximum likelihood.
4. Extend to multiple and multinomial logistic regression, and recognise confounding among predictors.
5. Contrast the discriminative (logistic) and generative (LDA/QDA/Naive Bayes) approaches to classification.
6. Compare classification methods and know when each is preferred.

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1. The sigmoid (logistic) function

$\sigma(z) = \frac{e^z}{1 + e^z} = \frac{1}{1 + e^{-z}}$ maps any real number to $(0, 1)$; it is monotone increasing, and $\sigma(z) \rightarrow 0$ as $z \rightarrow -\infty$, $\sigma(z) \rightarrow 1$ as $z \rightarrow +\infty$.

2. Odds and log-odds (logit)

For a probability p , the **odds** are $\frac{p}{1-p}$ and the **log-odds** (logit) are $\log\left(\frac{p}{1-p}\right)$. The logit is the inverse of the sigmoid: it maps $(0, 1)$ back onto all of $\mathbb{R}$.

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3. Likelihood and maximum likelihood estimation

Given data assumed to come from a distribution with unknown parameters θ , the **likelihood** $L(\theta)$ is the probability of the observed data as a function of θ . **MLE** chooses $\hat{\theta} = \arg \max_{\theta} L(\theta)$.

4. Bayes' theorem

$$P(Y = k \mid X = x) = \frac{P(X = x \mid Y = k) P(Y = k)}{P(X = x)}.$$

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A Classification Example: Credit Default

Setup

Predict whether a customer will default on their credit card debt, using predictors **balance** and **income**.

$$Y = \begin{cases} 1 & \text{Default} \\ 0 & \text{No Default} \end{cases}$$

The natural question

Can we just fit $Y = \beta_0 + \beta_1 X$ by least squares, the way we did for a quantitative response?

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Why Not Linear Regression?

Fit the coded response directly

Suppose we regress the 0/1-coded Y on X linearly: $Y = \beta_0 + \beta_1 X$.

- ▶ Predicted values can exceed 1.
- ▶ Predicted values can be negative.
- ▶ Fitted values cannot be interpreted as probabilities.
- ▶ The fit is sensitive to class imbalance and to how classes are coded numerically.

What we actually need

A model whose output is *always* a valid probability in $[0, 1]$, and whose predictions do not depend on an arbitrary numeric coding of the classes.

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Logistic Regression: The Goal

Model the probability, not the label

Instead of predicting $Y \in \{0, 1\}$ directly, logistic regression predicts

$$p(X) = P(Y = 1 \mid X).$$

Requirements on $p(X)$

- ▶ Must lie between 0 and 1.
- ▶ Must vary smoothly with X .
- ▶ Should be easy to turn into a class decision.

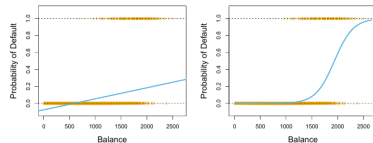


FIGURE 4.2. Classification using the `Default` data. Left: Estimated probability of `default` using linear regression. Some estimated probabilities are negative! The orange ticks indicate the 0/1 values coded for `default` (No or Yes). Right: Predicted probabilities of `default` using logistic regression. All probabilities lie between 0 and 1.

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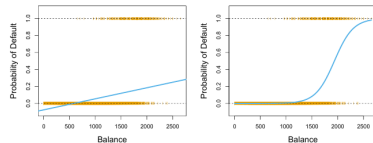


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4.3.1 The Logistic Function

Definition

Let $p(X) = P(Y = 1 \mid X)$. Logistic regression models it as

$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}} = \sigma(\beta_0 + \beta_1 X). \quad (4.1)$$

- ▶ Always between 0 and 1, for *any* β_0, β_1 .
- ▶ Smooth, monotone, *S*-shaped in X .
- ▶ As $X \rightarrow +\infty$, $p(X) \rightarrow 1$; as $X \rightarrow -\infty$, $p(X) \rightarrow 0$.

Decision rule

Classify as $Y = 1$ if $p(X) > 0.5$ (a common but adjustable threshold), else $Y = 0$.

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From Probability to Odds and Log-Odds

Rearranging Eq. (4.1)

$$\frac{p(X)}{1 - p(X)} = e^{\beta_0 + \beta_1 X} \quad \Rightarrow \quad \log\left(\frac{p(X)}{1 - p(X)}\right) = \beta_0 + \beta_1 X. \quad (4.2)$$

The key reinterpretation

Logistic regression is *linear in the log-odds*, even though it is non-linear in $p(X)$. A one-unit increase in X changes the log-odds by β_1 , i.e. it **multiplies** the odds by e^{β_1} .

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4.3.2 How Do We Fit β_0, β_1 ?

Linear regression

- ▶ Least squares.
- ▶ Minimises $\text{RSS} = \sum_i (y_i - \hat{y}_i)^2$.

Logistic regression

- ▶ Maximum likelihood estimation (MLE).
- ▶ Finds parameters that make the *observed* data most likely.

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Maximum Likelihood Estimation

Per-observation probability

$$p_i = P(Y_i = 1 \mid X_i) = \sigma(\beta_0 + \beta_1 X_i).$$

Likelihood function

$$L(\beta_0, \beta_1) = \prod_{i=1}^n p_i^{y_i} (1 - p_i)^{1-y_i}. \quad (4.3)$$

Fitting criterion

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg \max_{\beta_0, \beta_1} L(\beta_0, \beta_1) \equiv \arg \max_{\beta_0, \beta_1} \sum_{i=1}^n [y_i \log p_i + (1 - y_i) \log(1 - p_i)].$$

The right-hand side is the log-likelihood; maximising it has no closed form here and is solved numerically (iteratively reweighted least squares / gradient-based methods).

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Interpreting the Coefficients

Recall

$$p(X) = \sigma(\beta_0 + \beta_1 X).$$

Sign and magnitude of β_1

- ▶ $\beta_1 > 0$: probability increases with X .
- ▶ $\beta_1 < 0$: probability decreases with X .
- ▶ Larger $|\beta_1|$: stronger effect of X .

Role of β_0

- ▶ Baseline tendency of the model.
- ▶ Shifts where the S -curve crosses $p = 0.5$.

Odds-scale interpretation

A one-unit change in X multiplies the odds $\frac{p(X)}{1 - p(X)}$ by e^{β_1} — this is the natural, directly interpretable quantity to report.

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4.3.4 Multiple Logistic Regression

One predictor is rarely enough

Balance, income, age, education can *all* carry information about default risk. Extend the linear predictor to p variables:

$$P(Y = 1 \mid X) = \frac{e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}}{1 + e^{\beta_0 + \beta_1 X_1 + \dots + \beta_p X_p}}. \quad (4.4)$$

Same interpretive shift as multiple linear regression

β_j is the effect of X_j on the log-odds of $Y = 1$, **holding all other predictors fixed**. This matters, because it can reverse the story told by a single-predictor model.

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The Confounding Issue

A single-predictor model can mislead

Fit $P(\text{default} = 1 \mid \text{student})$ alone: being a student appears to **increase** default risk.

Add balance to the model

Fit $P(\text{default} = 1 \mid \text{student}, \text{balance})$: the coefficient on student **flips sign** — students tend to carry higher balances, and it is the balance that drives default risk, not being a student *per se*.

Lesson

Confounding: a predictor correlated with an omitted variable can take credit for that variable's effect. As with multiple linear regression, always ask “effect of X_j holding others fixed,” and fit the multivariable model before trusting any single-predictor story.

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4.3.5 Beyond Two Classes: Multinomial Logistic Regression

Binary case

$Y \in \{0, 1\}$.

- ▶ Spam / Not spam
- ▶ Default / No default
- ▶ Pass / Fail

$K > 2$ classes

$Y \in \{1, 2, \dots, K\}$.

- ▶ Digit recognition (0–9)
- ▶ Animal classification
- ▶ Sentiment: positive / neutral / negative

Idea

Compute a probability for every class and assign the observation to the class with the highest probability.

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Choose a reference class K

For each non-reference class $k = 1, \dots, K - 1$, model the log-odds against the reference:

$$\log\left(\frac{P(Y = k | X)}{P(Y = K | X)}\right) = \beta_{k0} + \beta_{k1}X_1 + \dots + \beta_{kp}X_p. \quad (4.5)$$

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Equivalent symmetric form

$$P(Y = k \mid X) = \frac{e^{\beta_{k0} + \beta_k^\top \mathbf{x}}}{\sum_{l=1}^K e^{\beta_{l0} + \beta_l^\top \mathbf{x}}}, \quad k = 1, \dots, K. \quad (4.6)$$

- ▶ Every probability is positive.
- ▶ All K probabilities sum to 1.
- ▶ Setting $K = 2$ recovers ordinary logistic regression — the softmax is the multi-class generalisation of the sigmoid.

Equivalent symmetric form

$$P(Y = k \mid X) = \frac{e^{\beta_{k0} + \beta_k^\top \mathbf{x}}}{\sum_{l=1}^K e^{\beta_{l0} + \beta_l^\top \mathbf{x}}}, \quad k = 1, \dots, K. \quad (4.6)$$

- ▶ Every probability is positive.
- ▶ All K probabilities sum to 1.
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SECTION 7

Generative Models: Bayes' Theorem, LDA, QDA

4.4 Discriminative vs Generative Classifiers

Discriminative (logistic regression)

Models the posterior directly:

$$P(Y | X).$$

Generative

Models the class-conditional density $P(X | Y)$ and the class prior $P(Y)$, then combines them with Bayes' theorem:

$$P(Y | X) = \frac{P(X | Y) P(Y)}{P(X)}.$$

Examples we cover

Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), and Naive Bayes.

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4.4.1 Linear Discriminant Analysis: Setup

Class-conditional densities

With K classes, let $f_k(x) = P(X = x | Y = k)$. Bayes' theorem gives

$$P(Y = k | X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^K \pi_l f_l(x)},$$

where $\pi_k = P(Y = k)$ is the prior probability of class k .

Classification rule

Assign x to the class with the largest posterior probability.

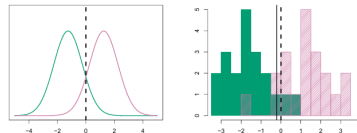


FIGURE 4.4. Left: Two one-dimensional normal density functions are shown. The dashed vertical line represents the Bayes decision boundary. Right: 20 observations were drawn from each of the two classes, and are shown as histograms. The Bayes decision boundary is again shown as a dashed vertical line. The solid vertical line represents the LDA decision boundary estimated from the training data.

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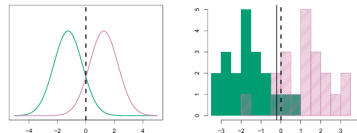


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LDA Assumptions for $p = 1$

Gaussian class-conditional density

LDA assumes $X \mid Y = k \sim N(\mu_k, \sigma^2)$: each class has its own mean μ_k , but *all classes share the same variance* σ^2 . Then

$$f_k(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu_k)^2}{2\sigma^2}\right).$$

- ▶ μ_k : class-specific mean.
- ▶ σ^2 : common variance across classes.

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LDA Discriminant Function for $p = 1$

Plugging into Bayes' theorem

Taking logs of $\pi_k f_k(x)$ and dropping terms common to every class yields the **linear discriminant function**

$$\delta_k(x) = x \frac{\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k). \quad (4.7)$$

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LDA for Multiple Predictors ($p > 1$)

Multivariate Gaussian assumption

With $\mathbf{x} = (X_1, \dots, X_p)^\top$,

$$\mathbf{x} \mid Y = k \sim N(\mu_k, \Sigma),$$

with class-specific mean vector μ_k but a **common** covariance matrix Σ .

Discriminant function

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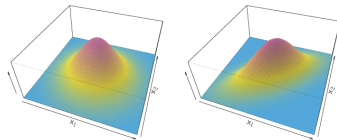


FIGURE 4.5. Two multivariate Gaussian density functions are shown, with $p = 2$. Left: The two predictors are uncorrelated. Right: The two variables have a correlation of 0.7.

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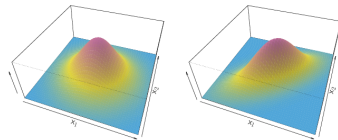


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Bayes Decision Boundary

Where two classes tie

Assign x to class k if $\delta_k(x) > \delta_j(x)$ for all $j \neq k$. The **decision boundary** is the set of points where two discriminant scores are equal, e.g. $\delta_1(x) = \delta_2(x)$.

For LDA, the boundary is linear

Because each δ_k is linear in x , the boundary separating any two classes is a hyperplane.

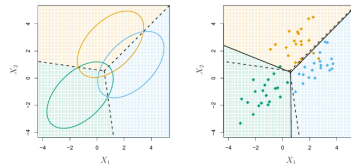


FIGURE 4.6. An example with three classes. The observations from each class are drawn from a multivariate Gaussian distribution with $p = 2$, with a class-specific mean vector and a common covariance matrix. Left: Ellipses that contain 95 % of the probability for each of the three classes are shown. The dashed lines are the Bayes decision boundaries. Right: 20 observations were generated from each class, and the corresponding LDA decision boundaries are indicated using solid black lines. The Bayes decision boundaries are once again shown as dashed lines.

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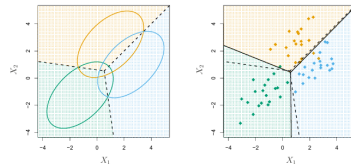


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Limitations of LDA

The key assumption

$$\Sigma_1 = \Sigma_2 = \dots = \Sigma_K = \Sigma.$$

All classes share the same covariance $\Rightarrow$ linear boundaries, a relatively simple model.

But real data often disagrees

- ▶ Different classes may have different variances.
- ▶ Different classes may have different covariance structures.
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Need a more flexible classifier $\Rightarrow$ QDA

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4.4.3 Quadratic Discriminant Analysis

Relax the common-covariance assumption

Instead of one shared Σ , QDA allows a **separate** covariance matrix Σ_k per class:

$$\mathbf{x} \mid Y = k \sim N(\mu_k, \Sigma_k).$$

Consequences

- ▶ More flexible: each class gets its own covariance.
- ▶ Decision boundaries become **quadratic** (curved).
- ▶ More parameters to estimate $\Rightarrow$ higher variance.

4.4 Generative Models for Classification 157

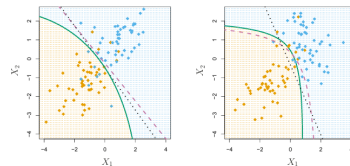


FIGURE 4.9. Left: The Bayes (purple dashed), LDA (black dotted), and QDA (green solid) decision boundaries for a two-class problem with $\Sigma_1 = \Sigma_2$. The shading indicates the QDA decision rule. Since the Bayes decision boundary is linear, it is more accurately approximated by LDA than by QDA. Right: Details are as given in the left-hand panel, except that $\Sigma_1 \neq \Sigma_2$. Since the Bayes decision boundary is non-linear, it is more accurately approximated by QDA than by LDA.

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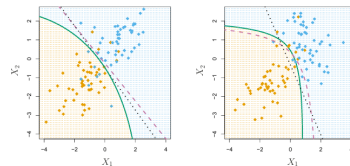


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4.4.4 Naive Bayes

Motivation

LDA and QDA model the *joint* distribution of all predictors. With many predictors, estimating a full covariance matrix (or matrices) becomes hard.

The naive simplification

Assume predictors are conditionally independent given the class: for $\mathbf{x} = (X_1, \dots, X_p)$,

$$P(\mathbf{x} \mid Y = k) = P(X_1 \mid Y = k) P(X_2 \mid Y = k) \cdots P(X_p \mid Y = k). \quad (4.9)$$

Payoff

Each $P(X_j \mid Y = k)$ can be estimated one dimension at a time — far cheaper than a full $p \times p$ covariance matrix, at the cost of (usually false) independence.

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Bayes' theorem plus independence

$$P(Y = k | X) = \frac{P(X | Y = k)P(Y = k)}{P(X)} \propto P(Y = k) \prod_{j=1}^p P(X_j | Y = k). \quad (4.10)$$

Classification rule

$$\hat{Y} = \arg \max_k P(Y = k) \prod_{j=1}^p P(X_j | Y = k).$$

Choose the class with the largest posterior probability.

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Comparison of Classification Methods

Method	Main assumption	Boundary
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LDA	Gaussian classes, common covariance	Linear
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Logistic Regression $\longrightarrow$ LDA $\longrightarrow$ QDA

General trend

Increasing model flexibility (LDA $\rightarrow$ QDA) comes at the cost of estimating more parameters — the familiar bias–variance trade-off, now in a classification setting.

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In Practice: Fitting Logistic Regression

```
import numpy as np
from sklearn.linear_model import LogisticRegression

# X: (n, p) predictors, y: (n,) in {0,1}
clf = LogisticRegression().fit(X, y)

# P(Y=1 | X) for new points
probs = clf.predict_proba(X_new)[: , 1]

# class labels using 0.5 threshold
labels = clf.predict(X_new)

print(clf.intercept_ , clf.coef_)
```

What changed vs. linear regression

LogisticRegression fits by maximum likelihood, not least squares, and exposes `predict_proba` for the probability itself, not just a hard label.

Habit

Inspect the predicted *probabilities*, not just the labels — the threshold (default 0.5) can and should be tuned to the problem's costs of false positives vs. false negatives.

Summary & What Comes Next

Key takeaways

- ▶ Classification predicts categorical outcomes; linear regression cannot enforce probabilities in $[0, 1]$.
- ▶ Logistic regression models $P(Y | X)$ directly via the sigmoid, fit by maximum likelihood; coefficients act on the log-odds scale.
- ▶ LDA, QDA and Naive Bayes are **generative**: they model $P(X | Y)$ and combine it with Bayes' theorem. LDA gives linear boundaries, QDA quadratic, and flexibility trades off against variance.

Next session: Resampling Methods

Every quantity we have reported so far — R^2 , coefficients, classification accuracy — was computed on the *training* set. How do we reliably estimate a model's **test** error?

Cross-validation and the **bootstrap** let us do this without a separate, expensive hold-out experiment for every model we consider.

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Classification

From categories to probabilities: logistic regression and generative classifiers

Questions & discussion welcome